Navigation2 Multi-Point Navigation

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1. Course Content

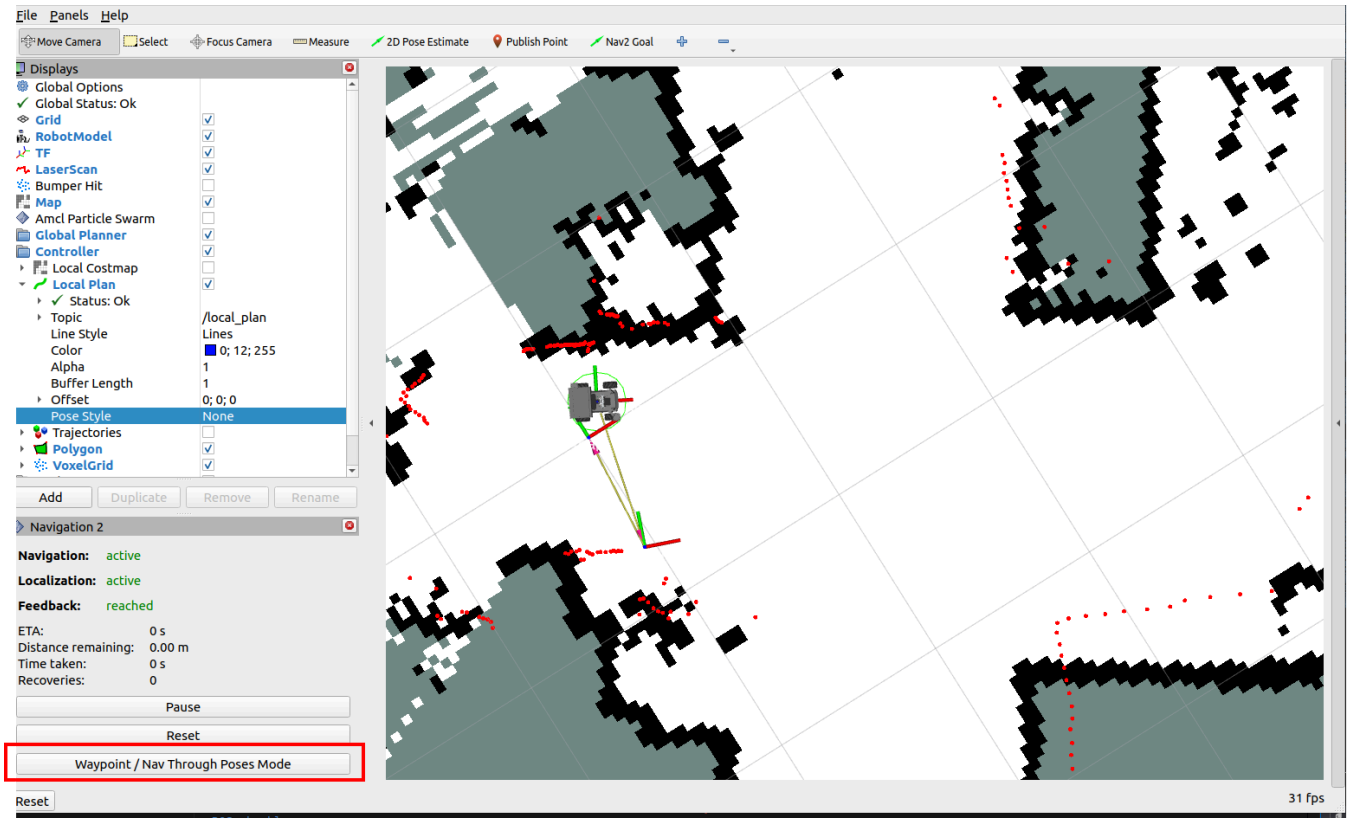
- Run the example program, manually mark multiple waypoints to achieve sequential navigation and obstacle avoidance between multiple marked points.

2. Preparation

- Referring to the prerequisite course [Navigation2 Single Point Navigation], first start the Nav2 navigation function and initialize the robot's position in the map

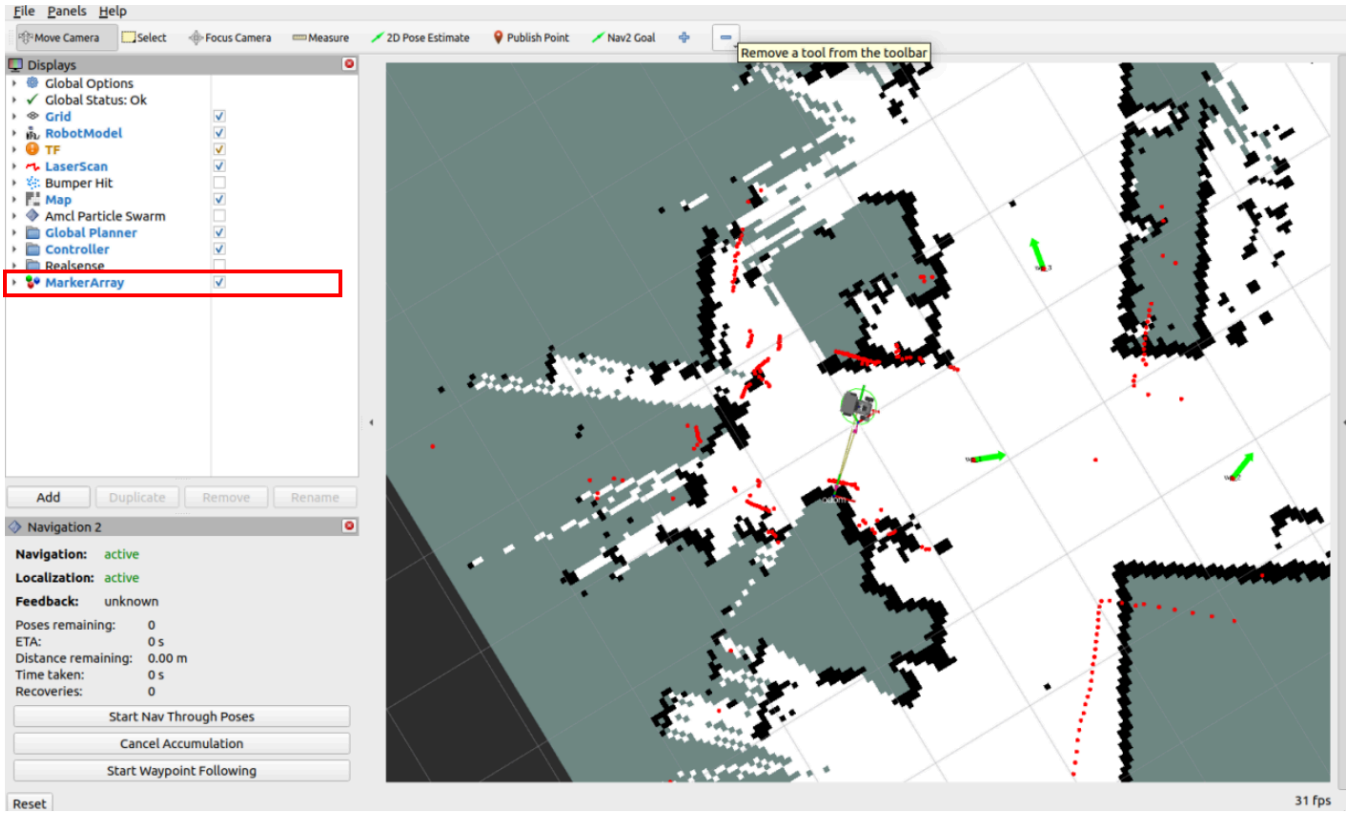
3. Running the Cases

- Click **[Waypoint/Nav Through PoseMode]** in the lower-left corner to enter multi-point navigation mode

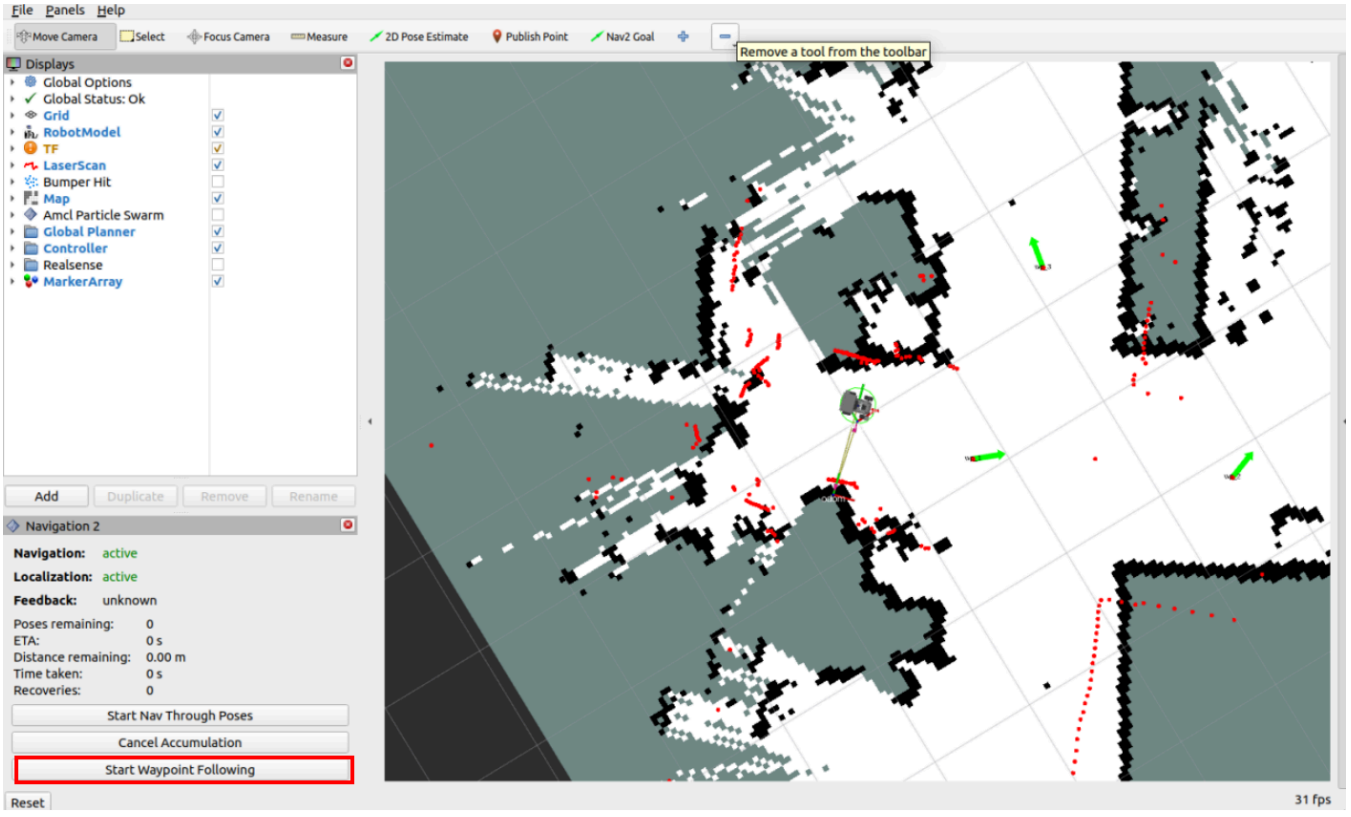


- Check **MarkerArray** in the left option bar to turn on the marker display, then click **Nav2 Goal** and use

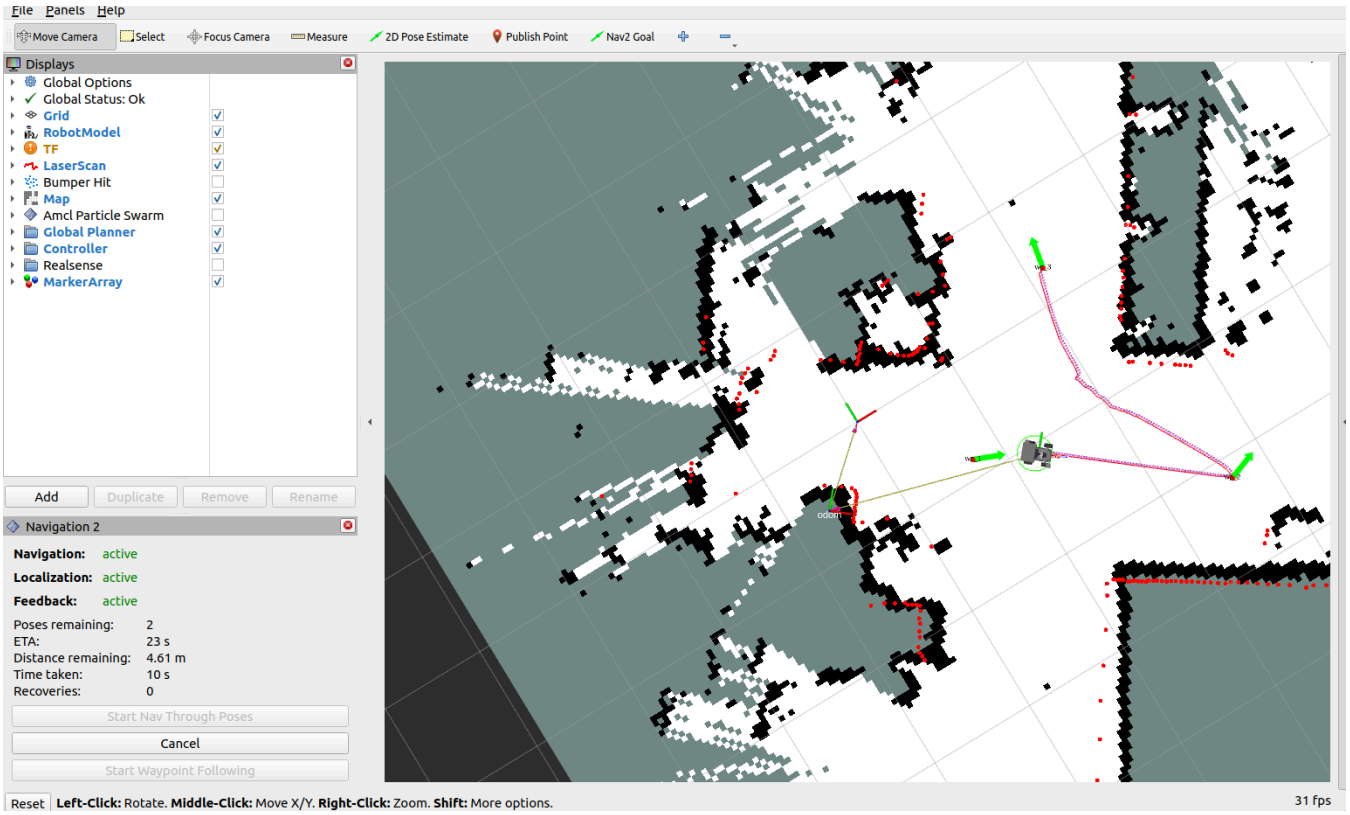
the mouse to mark multiple goal points on the map



Click **Start Waypoint Following** in the lower-left corner to start multi-point navigation



- The robot car navigates to the marked points one by one in order



4. Principle Analysis

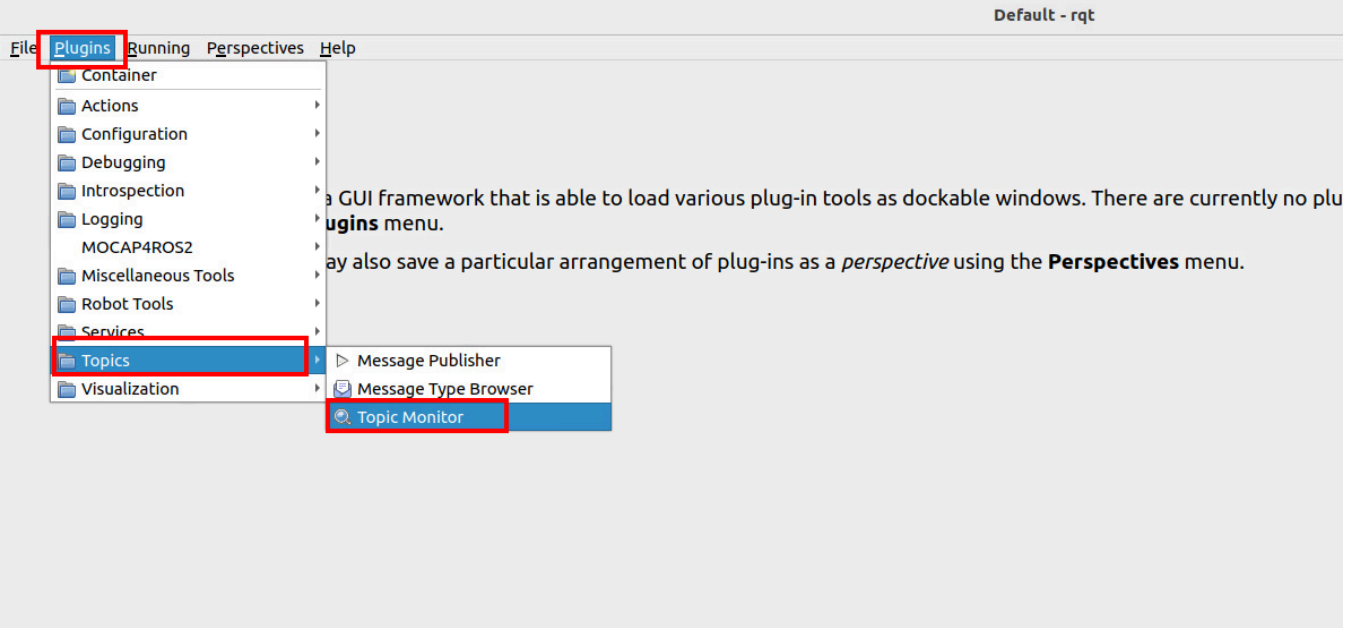
4.1 Viewing Waypoint Data

After the user opens **[Waypoint/Nav Through PoseMode]** and enters multi-point navigation mode, the point information marked by the user will be published to the **/waypoints** topic (the rviz waypoint navigation plugin adds some extra waypoints along the path between goal waypoints to smooth the path). We can view this **waypoint data** through the rqt interface

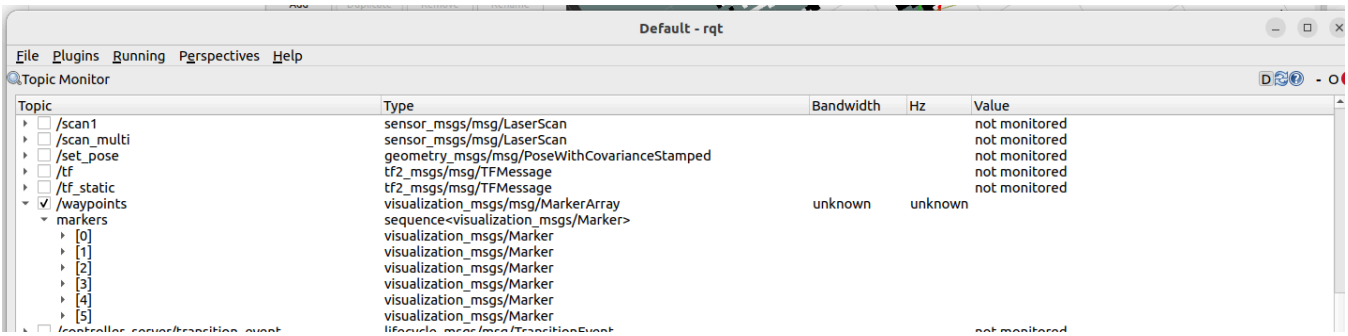
Startup command in the terminal:

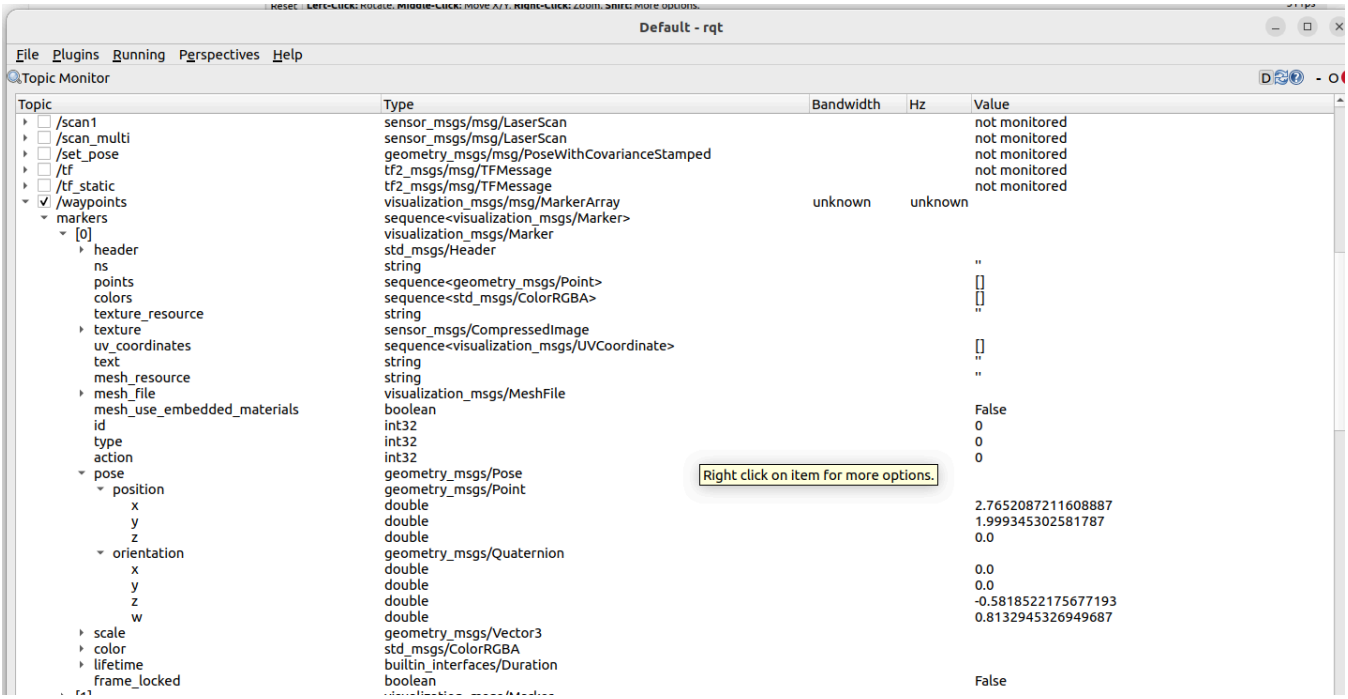
```
rqt
```

Click Plugins——>Topic——>Topic Monitor



- In the rqt interface you can see the **/waypoints** topic; after checking it, you can observe the data on the topic (you need to check the topic first, then publish waypoints in the rviz interface)
- The waypoints we manually mark in rviz will be published to this topic. Expand one of the waypoints to view the waypoint data; here we take [0] as an example, in which pose is the coordinate data





4.2 Data Sending and Execution

- After the waypoint coordinates are set, when you click **Start Waypoint Following**, the rviz plugin packages the waypoint coordinate sequence into a `FollowWaypoints` action request and sends it to the /follow_waypoint action server, which executes all the waypoints one by one
- View the node graph

```
rqt_graph
```

- In the node graph you can see the **/follow_waypoint** action server; this action server receives the waypoint sequence and then navigates to them one by one.

